Fault Code: 441

Battery Voltage - Out of Range Low

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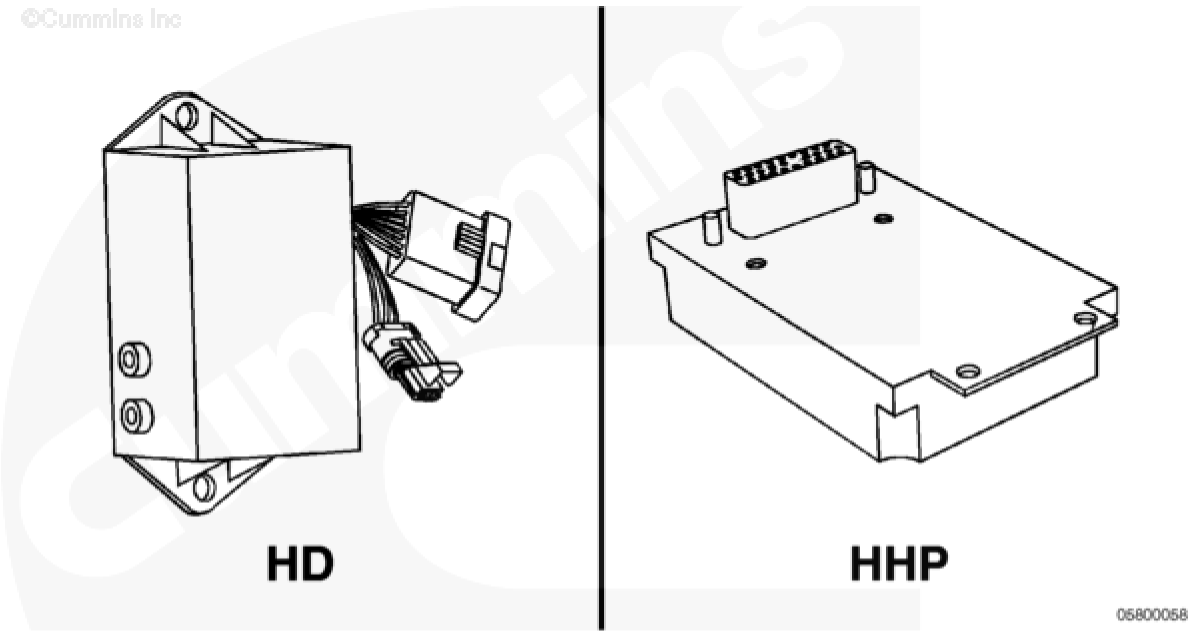
Overview

Codes	Reason	Effect
Fault Code: 441 PID(P): SPN: FMI: Lamp: Red SRT:	Battery Voltage - Out of Range Low. Battery voltage is less than the low threshold detected by the Centinel™ control module.	The Centinel™ control module voltage supply approaching a level at which unpredictable operation will occur.



Circuit Description

The Centinel™ control module receives unswitched battery voltage from the starter for heavy-duty. For high-horsepower, the Centinel™ control module receives switched power from the fuel shutoff valve solenoid. There are two in-line 5-amp fuses in the Centinel™ control module heavy-duty harness to protect the Centinel™ control module. The high-horsepower has **only** one 5-amp fuse. The battery return wires in the engine harness are connected to the engine block ground.

Component Location

The location of the battery will vary with the OEM. Refer to the OEM manual for the battery location.

Shoptalk

This fault can be caused by loose or corroded battery connections.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the equipment battery system.	
	STEP 1A. Inspect the battery cable connections.	No damaged connections
	STEP 1B. Check the battery voltage.	Heavy-duty: 9 to 32 VDC High-horsepower (12-VDC): 8.2 - 17.3 VDC High-horsepower (24-VDC): 15.5 - 30.3 VDC
	STEP 1C. Check the Centinel™ control module battery voltage.	Heavy-duty: 9 to 32 VDC High-horsepower (12-VDC): 8.2 - 17.3 VDC High-horsepower (24-VDC): 15.5 - 30.3 VDC
STEP 3.	Clear the fault code.	
	STEP 3A. Disable the fault code.	Fault Code 441 inactive

STEP 1. Check the equipment battery system.

STEP 1A. Inspect the battery cable connections.

Conditions :

- Turn the keyswitch OFF.

Action	Specification/Repair	Next Step

	No damaged connections	**1B**
• Corrosion • Loose connection.	**Repair or replace the damaged connections** • Repair the battery or starter connections. • Replace the battery or starter connections. Refer to the OEM service manual.	**2A**

STEP 1B. Check the battery voltage.

Conditions :
- Turn the keyswitch ON.

Action	Specification/Repair	Next Step
• Measure the battery voltage.	Heavy-duty: 9 - 32 VDC High-horsepower (12VDC): 8.2 - 17.3 VDC High-horsepower (24VDC): 15.5 - 30.3 VDC	**1C**
	Replace the battery Refer to OEM service manual.	**2A**

STEP 1C. Check the Centinel™ control module battery voltage.

Conditions :
- Turn the keyswitch ON.

Action	Specification/Repair	Next Step

• Heavy-duty: Measure the voltage between pins 1 (+) and 2 (-) of the Centinel™ control module harness connector. • High-horsepower: Measure the voltage between pins 22 (+) and 25 (-) of the Centinel™ control module connector. Refer to the wiring diagram for connector pin identification.	Heavy-duty: 9 - 32 VDC High-horsepower (12VDC): 8.2 - 17.3 VDC High-horsepower (24VDC): 15.5 - 30.3 VDC	**Complete**
	Replace the Centinel™ wiring harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	**2A**

STEP 2. Clear the fault code.

STEP 2A. Disable the fault code.

Conditions :

- Connect all the components.

Action	Specification/Repair	Next Step
• Connect all the components. • Start the engine and let it idle for 1 minute. • Verify that Fault Code 441 is inactive.	Fault Code 441 inactive	**Complete**
	Return to the troubleshooting step or contact your local Cummins Authorized Repair Location if all the steps have been completed and checked again.	**1A**

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